STATUS: WARNING

SOP-004: Predictive Maintenance - Thermal Alert Response

Effective Date: 2026-03-01 | Owner: Field Engineering | Review Date: 2026-09-01

1. Scope

This SOP applies to AI-generated thermal trend alerts from the TelcoPilot platform with a predicted hardware fault probability greater than 70%. It covers all tower sites across Lagos metro including Lagos West, Ikeja, Lekki, Victoria Island, Ikoyi, Apapa, and Festac.

2. AI Alert Trigger

An alert is triggered when the thermal sensor shows a consistent upward trend and the tower load exceeds 75% for more than 7 consecutive days.

3. Decision Matrix

Probability	Action	Timeframe	Owner
<70%	Monitor, check daily	N/A	System
70-85%	Schedule maintenance window	Within 4 hours	Field Engineering
>85%	Immediate action	Within 2 hours	Field Engineering + Manager

4. Immediate Action Steps

1. Schedule maintenance window and notify subscribers.
2. Pre-provision spare amplifier for hot-swap from depot.
3. Subscribe NOC to 80% thermal threshold alerts.
4. Execute hot-swap procedure if fault occurs.

5. Hot-Swap Amplifier Procedure

1. Power down affected amplifier.
2. Disconnect RF and power cables.
3. Remove faulty amplifier and install spare.
4. Reconnect all interfaces securely.
5. Power up and verify operational status.
6. Run continuity and thermal tests.
7. Update inventory and close maintenance ticket.

6. Tower Thermal Thresholds

Status	Temperature Range (C)

STATUS: WARNING

Normal	<65
Warning	65-72
Critical	>72

7. Spare Parts Inventory Requirement

Maintain a minimum of one spare amplifier per cluster to ensure rapid response capability.

8. Case Study Reference

Refer to Incident Report INC-2839 (TWR-LAG-W-031) where a thermal trend was detected 14 days prior to amplifier failure. Preventive action was recommended to avoid service disruption.

9. Version History

Version	Date	Change Summary
V1	2026-03-01	Initial release